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Professional avatars: librarians and educators in virtual worlds

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Abstract

Purpose – This study aims to examine librarianship and education in virtual world settings, focusing particularly on how librarians and educators establish professional identity and trust while teaching and providing information services as avatars within Second Life.

Design/methodology/approach – Data was gathered through semi-structured in-depth interviews with 12 librarians and educators within the virtual world of Second Life, and in visits to 50 virtual world workplaces.

Findings – Librarians and educators faced challenges of establishing professional identity and credibility for two different audiences – i.e. “newbies” and “residents” – within the “game-like” virtual world environment. Visual elements such as a conservative, humanoid avatar appearance, and imagery drawn from traditional workplaces were used by librarians and educators to establish their professional role with “newbies”, while for “residents” an upgraded avatar appearance was necessary to convey credibility.

Practical implications – For librarians and educators working within highly visual, graphical online settings such as virtual worlds, this study describes issues in providing avatar-mediated online distance education and online library services.

Originality/value – This research provides in-depth interviews and observations with librarians and educators from a variety of libraries and educational settings. The study also explores avatar-mediated communication, which is little-examined as yet in LIS research.

Keywords Virtual worlds, Second Life, Avatars, Digital reference, Distance education, Librarians, Distance learning

Paper type Research paper

1. Introduction

Today's librarians and educators stand at the crossroads of two worlds – teaching and answering questions within “brick and mortar” libraries and classrooms while also handling the “virtual branch” or “virtual classroom” for a growing audience of online users. In the USA alone, 3.9 million students enrolled in online courses in Fall 2007, increasing to 5.6 million by Fall 2009 (Allen and Seaman, 2008, 2010). Libraries support “virtual” users via e-mail, chat, instant messaging and other online technologies used among more than 12 million reference transactions handled by US research libraries during 2007-2008 (Kyrrilidou and Bland, 2009). Keeping up with new technologies in teaching and answering questions for online students is an ongoing challenge for librarians and educators.

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In exploring possibilities beyond text-based online communication, educators and librarians have experimented with virtual worlds such as Second Life (see www.secondlife.com). Second Life (SL) users enter a graphical, visually rich immersive environment in which the real-world self is represented by an online avatar that can speak, emote, gesture, and even rise into the air and fly. This virtual “self” is infinitely customizable and can appear as human, animal, mechanized, or an inanimate object.

As early as 2002, educators experimented with teaching classes in Second Life (Mon, 2009). Second Life Library 2.0, today known as the Community Virtual Library, launched in April 2006, and by 2007 librarians working as avatars on the Info Islands had answered 6,769 virtual reference questions (Alliance Library System, 2008; Bell *et al.*, 2007). Also by 2007, more than 70 colleges, universities, and schools had created an in-world presence (Jennings and Collins, 2007), establishing a building, teaching location, or campus.

Professional education work in a virtual world involves creating a personal avatar and learning to use it while teaching and communicating information. This article examines virtual world communication in educational contexts, seeking to understand the “professional as avatar” and its impact on the professional work of librarians and educators in virtual worlds.

2. Background

Most research on educators and librarians in Second Life involves descriptive case studies of classes (Delwiche, 2006; Jarmon *et al.*, 2008; Kemp, 2008; Sanchez, 2009) or case studies of individual libraries. (Buckland and Godfrey, 2010; Czarnecki, 2008; Godfrey, 2008; Hill and Lee, 2009; Swanson, 2007) As yet, few researchers have surveyed or interviewed educators and librarians working in virtual worlds (Luo, 2008; Luo and Kemp, 2008; New Media Consortium, 2007).

Luo (2008) surveyed 21 reference volunteer librarians about their work with the Second Life reference service, finding that the majority answered questions on Second Life topics, but fewer received non-Second Life related questions. Luo and Kemp (2008) surveyed 25 educators about teaching LIS students in Second Life, learning that the majority used Second Life for class activities, exercises and assignments rather than for lectures, and that many used the virtual world for group exercises and team-building, and as a place for students to give presentations. A survey of 209 educators, librarians and students by New Media Consortium (2007) found that 57 respondents reported teaching in the virtual world, and 103 stated that their institution either owned a virtual island or rented land in-world.

Two other methods used by researchers were comprehensive reviews of the research literature, and unobtrusive observations. Mon (2010) conducted an in-depth review of the research literature to provide a detailed account of Second Life teaching methods. Jennings and Collins (2007, p. 181) observed workplaces of 70 educational institutions in Second Life, finding that information was commonly communicated to visitors via visual displays such as signs (76.1 percent) or clickable web links (46.5 percent). The most comprehensive review of library workplaces was a sample of eight public libraries observed by Stimpson (2009), who identified potentially promising best practices such as offering library events, group memberships, gifts, and tutorials.

3. Purpose of study

Notably missing from the literature was exploration by LIS researchers of the impact of “avatar-mediated communication” (Castronova, 2003) on the professional work of librarians and educators. Second Life supports text chat, instant messaging and audio, and embodies librarians and educators as avatars with the ability to “recreate themselves” with any desired appearance as well as to modify their graphically rendered surroundings. This raises the key question: how does avatar-mediated communication influence professional information interactions conducted by librarians and educators in Second Life?

While some research has explored the impact of avatar self-representation on users (Boellstorff, 2008; Yee and Bailenson, 2007; Lee and Hoadley, 2006), only limited research has focused on educator and librarian avatar embodiment in a professional, working context (New Media Consortium, 2007). Improved understanding of the dynamics of professional identity in virtual worlds would support librarians and educators in more effectively utilizing highly graphical digital and virtual world environments for teaching and information services in future.

4. Method

This study utilized a qualitative, ethnographic approach of immersion in Second Life as a participant observer, followed by workplace observations and semi-structured in-world interviews with librarians and educators. Preparation began by joining Second Life in August 2006 as a participant observer, spending five or more hours per week for over a year and a half within the librarian and educator communities prior to the interview phase in May 2008. The participant observer phase included learning about librarians and educators in Second Life through direct experiences such as asking questions as a patron and answering questions as a librarian, attending classes as a student and teaching as an educator; and undertaking other roles in librarian and educator communities including audience member, group member, group leader, event organizer, meeting leader, conference attendee, conference planner, conference presenter, tour leader, tour follower, campus workplace builder and library builder. This immersive involvement facilitated preparation for interviews through acquiring functional skills and knowledge of the history, culture, and language of the virtual world’s users.

Twelve interviews were conducted from May 2008 to December 2008 with seven librarians and five educators either at their Second Life workplaces, or at an in-world research facility. During interviews, avatars sat on sofas or chairs around a recording device. The device recorded only participants who willingly touched the device to indicate their consent after being informed about the study and their rights as research participants.

Following one-hour semi-structured interviews, photographs were taken of avatars in their Second Life workplaces or at the research facility. Photos were also taken of librarian and educator workplace settings, and any available information objects were collected. Fifty in-world library workplaces were visited between May 2008 and January 2009.

Data analysis was conducted in two phases:

- (1) In Phase I, a first-level analysis explored how librarians and educators answered questions and provided information in Second Life (Mon, 2009). This

initial study described work activities and information tools used by librarians and educators, types of questions answered, how they provided information, and improved functionalities that they would have liked to have in better managing information work in Second Life.

- (2) In Phase II, the current study, a secondary-level analysis focused on the avatar-mediated aspect of librarians' and educators' professional work in Second Life – the visualized self-representations of librarians, educators, and their professional workplaces, and the impact of these visualized representations on information work and teaching. Emerging in this second phase of the study was an issue uniquely challenging to librarians and educators working within virtual worlds – establishing professional identity, credibility and trust while embodied as an avatar in a highly visual, “game-like” setting.

Data analysis was guided by a theoretical framework drawing upon the work of sociologist Erving Goffman on presentation of the self (Goffman, 1959). Goffman's work has influenced virtual world researchers including Carr (2008) in research with students in Second Life, Boostrom (2008) on virtual identity for new users in Second Life, Delwiche (2006, p.164) on identity within Everquest, and McDonough (1999) on virtual identity in research with designers of virtual worlds.

Goffman's work was grounded in symbolic interactionism and focused on impression management and presentation of the self within interpersonal interactions. (Blumer, 1969; Goffman, 1959) In Goffman's view, interpersonal interactions were performative, staged and framed within each participant's expectations about the setting, their socially-defined roles, and actions perceived as appropriate for the encounter. Individuals maintain multiple public roles or “selves”, selectively presenting the “self” appropriate for each audience (Goffman, 1961). Within this social performance of identity, one could “lose face” when not treated with appropriate respect during an encounter (Goffman, 1955). Individuals may selectively reveal or conceal information about the self, the “information preserve”, and seek to exert control over who can summon him or her into conversation, or the “conversational preserve” (Goffman, 1997a, p. 140; Goffman, 1997b, pp. 50-1). Goffman's conceptions of frame, setting, roles, performative identity, stigma, face, the information preserve and conversational preserve provided the framework for data analysis. Member-checking was used in discussing findings with study participants, and in presenting results and inviting comments at real world and in-world conferences and discussion groups attended by Second Life librarians and educators. It should be noted that the generalizability or transferability of these results is limited by the small sample size as well as the highly changeable nature of virtual environments over time. As will be discussed in the conclusions, virtual world builders sometimes introduce key modifications to these environments, and particularly in response to findings of researchers.

5. Findings

5.1 *The professional as avatar*

Any self-representation that can be imagined can be created in Second Life. Asked about unusual avatars they had seen, librarians and educators described everything from animals and insects (“a giant beetle”, “a squirrel”, “a butterfly”) to monsters and imaginary creatures (“a tiny fairy”, “a dragon”, “a frog prince”, “a flying spaghetti

monster”) and inanimate objects (“a shiny, floating sphere”, “a box”, “glowing neon tubes”, “a puff of smoke”). Developing professional aplomb regarding user appearance is helpful. One example of matter-of-fact acceptance of unusual avatar self-representations was described by a librarian: “I once had a reference interview with an avatar who had a bloody sword piercing his abdomen [...] He never mentioned it, nor did I”.

Librarians and educators sought to craft their avatar self-representations in ways that would facilitate professional communication. A commonly expressed sentiment was, “I want an avatar that will be taken seriously”, and specific attributes mentioned included “humanoid”, “conservative”, “friendly and professional”. One educator made a distinction regarding appearance and different audiences: “avatars that don’t know me might not take me seriously when I am a monkey or a bear”.

Although in the real world (“real life” or “RL”), clothing has both utilitarian functions and performative aspects, virtual world clothing is almost entirely performative. To convey a serious, scholarly, and professional avatar self-representation, educators and librarians wore business suits, eyeglasses, and conservative clothing. For female avatars, simply finding professional clothing was a challenge, with bikinis and ball gowns more easily found than business suits. A librarian noted, “I did try to find some more business-like clothing in SL, but it was hard. Most clothing is rather extreme [...] low cut, party dresses, very short skirts, tight tops”. Second Life is perceived as a “game” by many users, making it unexpected to encounter working professionals within this context (Pence, 2007/2008). Although clothing stores were established early (Ondrejka, 2004), available designs primarily support entertainment. As another librarian commented, “I work to appear un-barbie-like”.

Goffman (1956, p. 269) mentions “audience segregation” in which a different self may be acted out for different audiences. For educators and librarians in Second Life, avatar and clothing choices were influenced by cultural expectations of two major audiences of virtual world users – “newbies”, who are the newer arrivals (sometimes derogatorily called “noobs”), and “residents”, who are more seasoned participants who “live” and “work” in Second Life, perhaps even renting or owning land within the virtual world.

A librarian commented about choosing a conventional and conservative humanoid appearance: “I didn’t want to unnerve anyone who was new to SL”. Another librarian said, “I wanted to be accessible, unthreatening to newcomers”. However, residents posed another challenge for professional self-presentation. Residents typically present a more “polished” avatar appearance due to significant time, effort and/or money invested into self-created or designer-made clothing, adornments, “photo-realistic” or elaborately rendered avatar skins and shapes, and animations enhancing gestures and movements such as a more natural-looking or more stylized walk. Since a resident is unlikely to seek information from a “newbie”, librarians and educators needed to provide both a traditional conservative humanoid appearance for newbies while also signaling to residents a sophisticated avatar appearance. Thus, an educator commented on his avatar appearance: “paid a good penny for it”, while a librarian described purchasing a higher-quality skin to better connect with residents: “I stuck with a noob skin for a long time [...] But my community is so sartorially obsessed, I finally decided it was leaving me out of an important cultural conversation with my main patron group”.

Gender could easily be modified by changing an avatar’s shape, skin, and hair. A librarian discussed choosing between female and male self representations based on

what might best fit expectations in a community of residents: “I have a basic female and basic male avatar... [the community] has an easier time with a nice bumbling male”. Another librarian described advantages of the female gender: “I have a male friend who uses a female alt at the ref desk because she is more ‘approachable’”. Ethnicity represented another performative choice. An educator said, “wherever I went I saw very few people of color [...] as avatars [...] I thought I’d do what His Holiness the Dalai Lama is reported to have said when asked what he’d come back as if he could choose his next incarnation. He reportedly said, ‘A woman of color’”.

Librarians and educators sometimes used their own avatars in informational and learning scenarios. One librarian spoke of morphing appearance and gender to embody a historic figure in a re-enactment. An educator discussed using different “selves” for case study learning scenarios in which students observed a scene, deduced what had occurred, and determined next steps. Here, the avatar becomes part of the informational and learning environment.

5.2 *Alternative identity*

Additional virtual “selves” are created by adding new accounts, or “alts”. Five of the librarians and educators reported maintaining “alt” identities separate from their “professional” avatars. Less effort was generally invested in professional self presentation for “alts”. A librarian commented, “My alt has blue hair”. Another librarian said, “I don’t spend lindens or time fixing that avatar up” (“Lindens”, the virtual-world currency, are purchased with real-world money). The “alt” identity was sometimes used in work as an actor in an educational video, or a role-playing character in a teaching scenario. Work shifted to alts included backing up important inventory items, conducting research, and managing money.

Alternate identity afforded a respite from being summoned into conversations. A librarian noted, “I started using another one [alt] just so I can come in and work in privacy”. The main avatar was “associated with all my inventory and contacts” but those same contacts could see whenever the librarian or educator logged in, sending instant messages (IMs) which interrupted planned work. Another librarian commented: “I am considering using [the alt] to work sometimes because I find now that I get so many IM’s while trying to work!”.

However, the “alt” generally was not considered the “real” avatar self. An educator described the main avatar: “this one is ‘me’”. A librarian commented, “I feel weird coming inworld as my alt [...] it is like pretending!”. Identification with the “main” avatar sometimes included imbuing it with elements of real-life appearance. One librarian said, “I made my avatar look like my RL self”. An educator said, “I wanted it to look like myself as I knew I was going to be doing work here”. Only one study participant consistently used distancing language, speaking of the main avatar in third person: “I don’t really worry about his appearance too much” and referring to the avatar by name: “After a few moments of introspection and reflection, I changed [avatar name] back”. Other study participants placed greater emphasis on their main avatar’s appearance and role. As one educator stated, “my SL avatar is an extension of me into the virtual world”.

5.3 *Dynamics of personal and working spaces*

Just as the avatar was infinitely customizable, workplace surroundings could be customized to frame encounters. Librarians and educators often chose visual

metaphors of a university classroom, campus, or library. Surrounding imagery of reference desks, classrooms, books, bookshelves and computers influenced expectations about professional roles and appropriate contextual behaviors.

In observations of 50 libraries in Second Life, commonly observed workplace features included photos and posters (80 percent), books and bookshelves (68 percent), computers (56 percent), and reference desks (20 percent). A strong identification of libraries with books was seen not only in workplace imagery of books and bookshelves, but also in a librarian's comment that a commonly asked patron question was, "Where are the books?". For institutions within role-playing or themed communities (e.g. ancient Rome, the American Old West), workplace imagery might include a papyrus scroll instead of a book, or a steam-powered card catalog rather than a computer, but generally still conveyed scholarly purpose. While some virtual workplaces included books that could be read and computers that opened web pages when touched, in others a bookshelf or laptop served only as a non-functioning visual backdrop. The workplace setting contributed to, as one librarian put it, "the crucial importance of establishing trust". A commonly reported user response among librarians was "people asking "are you really a librarian?" and "surprise at the fact that I am a 'real' librarian".

While some librarians and educators identified specific settings in which they worked (e.g. "an open courtyard with walls that are used for displays, and a labeled desk area") others described ways of working not tied to any single place. An educator said, "There is no consistent setting". Librarians spoke of doing their work "anywhere" and "everywhere" without the need for a specific location.

The virtual world environment offered possibilities for conducting information interactions beyond traditional boundaries and familiar replications of real-world workplaces. An educator said, "I think it is just the beginning with folks going to what they are familiar with and then moving them to things and places that are not like real life and to places and things we cannot do in RL due to time and distance". One example that librarians and educators described of breaking boundaries was real world limitations to serving only geographically local users. Within the virtual world, clientele changed by time of day – over 24 hours, the world came into the workplace with changing languages and timeframes of activity marking different nationalities of incoming users. Librarians designed information resources for an international clientele ("basic SL info in a couple of languages") and carried translation tools for patrons speaking other languages.

Both educators and librarians questioned rebuilding traditional settings in the virtual world environment. Educators spoke of teaching in open spaces such as "sandboxes" with plenty of room for students' avatars. An educator commented: "a classroom is a classroom in RL or SL, and can be just as confining". Librarians talked about departing from traditional library workplaces by placing reference desks outside of buildings, or tearing away walls and ceilings. A librarian noted: "When I was providing volunteer reference service, someone created a very futuristic looking reference desk. The funny thing was, no one ever used it. Not librarians. Not patrons". Changing aesthetics, such as a garden of giant flowers replacing a botany laboratory, offer new challenges in asserting professional identity within workplaces not immediately recognizable as classrooms or libraries.

6. Conclusions

This study found that librarians and educators faced the problem of commonly held perceptions of the virtual world as a game rather than a place for education and learning, a difficulty also discussed by Carr (2008) who referenced Goffman's *Frame Analysis* (Goffman, 1974) in terms of how a game-like environment could influence students' understandings, or misunderstandings, of the serious nature of in-world teaching and learning activities.

In reframing the virtual world as an environment for professional activity, librarians and educators "set the scene" with workplace visual elements emphasizing educational purpose such as computers, books, and bookshelves. Stimpson (2009, p. 16) similarly found that computers were a typical workplace feature for in-world public libraries. However, traditional design motifs are observed beyond educational settings. Sociologist Nathaniel Glazer (2005) wrote of his surprise upon seeing throughout Second Life how "conservative and traditional the created spaces of the virtual world are".

For an avatar appearance that would be 'taken seriously', librarians and educators often chose traditional, conservative humanoid self-representations. Other research on avatars in business and sales confirms humanoid appearance as more "liked" by customers (Keeling *et al.*, 2010). Librarians and educators purchased items for a higher-quality appearance in deference to expectations of Second Life residents, which echoed findings in research by Boostrom (2008, p. 17) on how purchased items enhanced identity for stigmatized newbies. Several other studies (Taylor, 1999; Messinger *et al.*, 2008; Yee and Bailenson, 2007) emphasize the importance of attractive avatar appearance. In research by Taylor (1999, p. 441) one participant summed up the impact of avatar identity as performative choice: "people treat you based on how you present yourself, and, if you pay attention, you'll notice that *you* change depending on how you present yourself". Second Life has recently responded to researcher-identified stigmatization of new users by introducing a key modification – regularly changing, higher-quality avatars for incoming users, making it more difficult to visually identify new users.

Alternative identity was sometimes used in creating a separate "self" for fulfilling tasks, with five of the librarians and educators (42 percent) reporting that they maintained an "alt" identity. The New Media Consortium (2007, p. 5) found that among 207 educators and librarians surveyed, 33 percent created two or more Second Life accounts. Alternative identity can extend to other virtual worlds as well. Taylor (1999, pp. 439-40) refers to "plural existence" in which users have digital selves in not just one, but multiple virtual worlds. Second Life is one among hundreds of virtual worlds in existence.

Although the extent of mainstream adoption of virtual worlds remains to be seen, some trends toward increasing use can be observed. During 2009, Second Life users logged 481 million in-world hours and spent \$US567m dollars on virtual goods and services, and in December 2009 traffic increased upon release of James Cameron's blockbuster movie *Avatar* (Linden, 2010a; Linden, 2010b). Open source virtual world software options coupled with recent price increases by Linden Labs sparked a diaspora to other virtual worlds such as ReactionGrid (see <http://reactiongrid.com/>). Meanwhile, game-based learning continues to gain traction among educators (Annetta *et al.*, 2010) as children grow up with online games and virtual worlds.

Traffic of children into virtual worlds is driven by corporations bundling virtual world logins with children's toys such as Webkinz stuffed animals, Bratz dolls and Barbie dolls. Millions of young users play and socialize together in Runescape, with 4.5 million users in 2009, Club Penguin, with 4.6 million unique visitors in March 2008 alone, and other settings (Buckleitner, 2008; Snider and Molina, 2009). Steinkuehler and Williams (2006, p. 886) described online games as potential "third places", while Crowe and Bradford (2006, p. 337) suggested that parents may perceive virtual public spaces as safer for children than physical public spaces. For young users, growing up with gaming and virtual worlds may raise future expectations of more visually engaging and immersive experiences within online libraries and online education.

A continuing challenge for librarians and educators is in evolving best practices for avatar-mediated interactions. One librarian in this study described "a hybrid of library and museum techniques" allowing "the interplay of words and images to tell the story" within virtual world workplaces. While it remains too soon to speak of universally-accepted best practices, the work of early adopters provides foundational groundwork toward establishing professional standards and new media literacies in librarianship and education within virtual worlds.

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